

What Is SIM-CONNECTED IoT?

The future of connected intelligence rests on trust, and that trust begins with secure, SIM-enabled foundations.



SACHIN ARORA
is the Head of
Connectivity and
IoT for India at
Giesecke+Devrient

Artificial intelligence (AI) and machine learning (ML) are transforming industries, unlocking new efficiencies and capabilities. Their integration with the Internet of Things (IoT) is among the most significant shifts, creating smarter, more responsive, and secure connected ecosystems.

IoT's evolution has accelerated, and it now supports real-time analysis of massive datasets at unprecedented scale. AI and ML push this further by turning IoT systems into intelligent, autonomous networks that enable sharper decisions, higher efficiency, and faster responsiveness.

As IoT ecosystems grow more complex, their performance increasingly hinges on one factor: the integrity of the data powering them. This integrity determines how much value AI can deliver. Technologies such as embedded SIM (eSIM), integrated SIM

(iSIM), blockchain, and elliptic curve cryptography (ECC) are strengthening this foundation, ensuring IoT systems remain intelligent and the vast data they generate remains trustworthy.

Market trends underline this rising focus on security. India's IoT security market was valued at US\$205.9 million in 2024 and is projected to surge to US\$2340.9 million by 2033, at a CAGR of 31 per cent, as organisations reinforce their systems.

Why secure, SIM-connected IoT is essential

When AI and IoT come together to predict equipment failures, optimise energy usage, or power autonomous vehicles, the benefits are substantial. However, with greater interconnectivity comes greater risk. Each connected device becomes a potential entry point for cyber threats, and any data compromise can lead to flawed AI-driven decisions.